H.J. Baker & Bro., LLC

AP Invoice Processing Platform

Solution Blueprint

An intake-to-payment platform that reads, classifies, matches and posts accounts-payable documents to Sage X3. leaving people to handle only the exceptions.

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1 Executive summary

H.J. Baker processes accounts payable for seven operating companies through a single mailbox and Sage X3. The current tool, Blue Creek, holds the document image and routes it for approval, but the keying, coding and matching all happen by hand in Sage afterwards. That manual effort is the largest cost in the process.

This platform automates the journey from arrival to payment. Every document that lands in the AP mailbox is read and classified by AI, matched against Sage, routed for approval within the right limits, and posted back to Sage with its image attached, with people stepping in only where a genuine decision is required. Automation is earned company by company as accuracy proves itself, so the business is never asked to trust more than the results support.

The platform removes five specific constraints that shape AP today: manual line coding, Blue Creek's 500-line ceiling, hourly file-based Sage integration on a 90-day window, a settlement chain that cannot be proven end to end, and the absence of operational visibility.

Table 1. *At a glance*

Replaces	Blue Creek (Vision360)
System of record	Sage X3. the platform never becomes a second ledger
Document types	PO, Non-PO, Prepayment, Credit, Rebill
Operating companies	Seven, across USD, CAD and GBP
Volume	Approximately 3,300 invoices per month (from the Sage export sample)
Headline targets	70%+ of PO invoices straight through to approval; under 15% of lines keyed by hand

2 Business objectives

The platform is measured against outcomes, not features. Each objective below has a measure and a target reported at the end of hypercare.

Table 2. *Objectives and success measures*

Objective	Measure	Target
Reduce manual coding	Lines requiring manual entry	Under 15% of all lines
Straight-through processing	PO invoices reaching approval with no human edit	70% or better
Classification accuracy	Documents routed to the correct queue without human correction	95% or better
Cycle time	Arrival to Sage document created, median	Under one working day
Settlement integrity	Rebill chains with all three Sage numbers linked	100% of settled rebills

Objective	Measure	Target
Nothing lost	Documents in a queue with no owner and no next action	Zero
Operational visibility	Volume, automation, exceptions and vendor detail on live data	Available and clickable to source

3 Scope

3.1 In scope

- Intake from the AP mailbox, including multi-attachment emails and scanned batches.
- AI extraction of header and line data with per-field confidence, and AI classification into five document types.
- The five processing flows in full, including every point where a person must decide.
- Two- and three-way matching against Sage purchase orders and receipts, and line coding from orders and templates.
- Approval hierarchy, limits, escalation, delegation and email approval; rule-driven AP final review.
- Creation of purchase invoices and credit memos in Sage with the image attached, document-number write-back, and payment feedback.
- Prepayment positions and drawdown; credit and rebill settlement chains.
- Search, dashboards, analytics and a complete audit trail; administration of routing rules, templates, reference lists and users.

3.2 Out of scope

- Purchase order creation, confirmed not automatable; the platform asks for the order number or processes a template-coded supplier invoice.
- Any change inside Sage X3, Sage licensing, and all Sage-side development, which sits with Net at Work.
- Requisitioning, goods receipting, inventory and the general ledger; payment execution and bank files.
- Vendor portal, vendor onboarding and statement reconciliation; accounts receivable, expenses and payroll.
- Commercial and hosting matters, effort, price, infrastructure and service levels are covered in the Statement of Work.

4 Solution overview

One route in, one classification decision, five destinations, one system of record. A document arrives, is read and classified, is routed to the queue for its type, is validated and (where relevant) matched against

Sage, is approved within the right limits, and is posted to Sage, which returns the document number and, later, the payment.

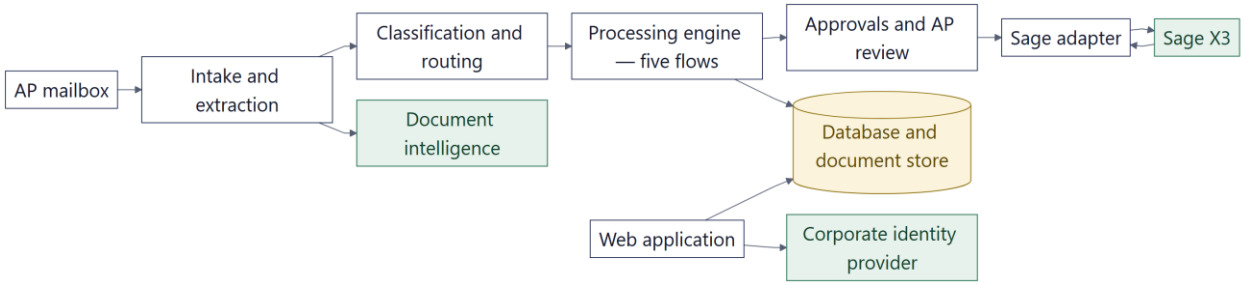


Figure 1. *The platform end to end: one intake path, one Sage boundary, one place state lives.*

■ platform component ■ data store ■ external service

Sage X3 remains the system of record throughout. The platform reads master data live at the moment it is needed, so it never works from a stale copy, and writes each document with its image attached in a single confirmed call.

Table 3. *The pipeline, stage by stage*

Stage	What happens	Automated
Intake	Ingest each attachment as a document, split batch scans, refuse duplicates	Yes
Extraction	Read header and every line with confidence per field	Yes
Classification	Route to one of five types by rule; below threshold, hold for a person	Yes, with a human fallback
Validation & matching	Confirm totals and fields; match against PO and receipts; code the lines	Yes; exceptions to a person
Approval	Route by hierarchy and limit; AP final review where a rule fires	Human decision
Post to Sage	Create the purchase invoice or credit memo with the image; capture the number	Yes
Payment	Capture payment status from Sage and close the document	Yes

5 Core capabilities

Ten capabilities carry the platform. Each is described below in terms of what it delivers to the business.

5.1 Document intake

Every document that arrives in the AP mailbox is ingested as its own record, with its source email preserved for later reading. Scanned batches of several invoices in one file are separated automatically, and each separated document is traceable back to the original file and page range.

- Accepts PDF, TIFF and common image formats, single or multi-page.
- Refuses duplicates on vendor, invoice number and amount together, naming the document that already holds the number.

5.2 AI extraction

Header fields and every line are read from the document with a confidence score for each field. There is no line-count ceiling: a load list running past a thousand lines is read whole. The vendor record is read live from Sage at the moment of extraction, so terms, currency and pay method always reflect Sage rather than a stale copy.

- Fields below a confidence threshold are marked for a person rather than accepted silently.
- The image is shown beside the extracted data with each field's confidence visible.

5.3 AI classification

Documents are classified into five types by a priority rule set: a prepaid vendor makes a prepayment, a negative total or cancellation wording makes a credit, a settlement vendor re-invoicing at a settled price makes a rebill, a valid purchase order makes a PO invoice, and everything else is a Non-PO invoice. The first rule that matches wins, and the rules are configuration the AP team owns.

- A single confidence threshold decides whether the platform routes on its own or holds the document in Manual Review with the reason stated.
- An unknown vendor lowers confidence, because that is exactly where a confident-looking guess is most dangerous.

5.4 Validation

Before a document can be submitted, its arithmetic and its key fields must hold. Net plus taxes plus freight must equal the stated gross, and the four key header fields, vendor, invoice number, date and amount, are confirmed against the image, so a supplier's default can never override what the document actually says.

- Every coding value is validated against the live Sage reference lists, so a code retired in Sage cannot be posted.
- A purchase order already consumed by another invoice cannot be attached again, and the holder is named.

5.5 PO matching

Purchase-order invoices are matched against Sage automatically. The match type, two-way or three-way, is read from the purchase-order line itself, not decided by the platform. Three-way matching is at

receipt level, keyed on the ship date and bill-of-lading pair, which is what makes a monthly summary covering a thousand receipts tractable.

- Quantity and price are checked within configured tolerance; outside tolerance becomes an exception showing the variance, not an automatic rejection.
- Matched lines are coded from the order, the manual coding that is the largest cost today.

5.6 Approval workflow

Every document is approved before it reaches Sage. Approval routes by hierarchy and per-company limit, escalating automatically when an amount exceeds an approver's limit, and nobody approves a document they submitted. A document matched automatically still requires approval unless the AP team has explicitly exempted the vendor or condition.

- Vendors on the hierarchy list route by hierarchy; everything else routes ad-hoc to the person the business names.
- A traveling approver can approve from email; delegation is time-boxed and audited.

5.7 Sage integration

The platform reads and writes Sage X3 live. Vendors, purchase orders, receipts and reference lists are read at the moment they are needed; purchase invoices and credit memos are created with the image attached in a single call that returns the Sage document number, which is written straight back onto the record.

- Writes are idempotent on the platform's own document number, so a retry after a timeout cannot post the same document twice.
- A daily reconciliation compares platform records against Sage documents and reports any divergence.

5.8 Payment tracking

Once Sage pays a document, the payment number, date and method are captured and shown on the record, and the document is closed. This replaces the hourly file feed and the 90-day lookback that limit visibility today.

- Payment method is recognised from the Sage payment prefix (ACH, wire, EFT, direct debit).
- Prepayment documents draw down against the open position, and the consumed balance is recalculated on each drawdown.

5.9 Search and audit

Every document is searchable across header and line fields, and every count shown anywhere in the platform, a dashboard tile, a queue badge, carries the exact filter that produced it and clicks through to those documents. A count you cannot click through to is treated as a defect.

- A complete, append-only audit trail records every state change, edit, approval and Sage write with timestamp and actor.

- Each document's own history reads from the document: when it arrived, when it was classified and at what confidence, and every step since.

5.10 Reporting

Dashboards answer the questions the business asked directly: how many documents were processed this month and quarter, how many were auto-matched versus keyed by hand, and how they split by PO, non-PO and vendor. Every figure is clickable through to the documents behind it, and every view exports to spreadsheet.

- Operational views cover volume, automation rate, cycle time, exceptions, approvals, prepayment positions and settlement-chain integrity.
- Vendor-level detail shows volume, exception rate and cycle time per vendor.

6 Document processing

The five document types are not a filing convention. Each has its own trigger, its own checks and the point at which it must stop for a person, and each produces a different result in Sage.

Table 4. *The five document types at a glance*

Type	Recognised by	Becomes in Sage	Always stops for	Distinctive rule
PO invoice	A valid purchase order on the document	Purchase invoice (PIN)	Approval	Match type read from the PO line; coded from the order
Non-PO invoice	No purchase order present	Purchase invoice (PIN)	The PO decision, then approval	A person attaches an order or codes from a template; the platform never creates the order
Prepayment	The vendor is marked as prepaid	A payment, then invoices drawing it down	Approval, usually CFO for the lump sum	Size alone never classifies; the document must say it is a prepayment
Credit	Negative total or cancellation wording	Credit memo (PCM)	The original invoice reference, then AP review, always	Must reference the original invoice it reverses; never posted as an invoice
Rebill	A settlement vendor re-invoicing at a settled price	Purchase invoice linked to the credit	The credit-memo reference, approval, AP if the period is closed	Must name the credit memo it settles, linking original, credit and rebill

7 Business rules

The rules that govern how documents are routed, checked and posted. They are configuration the AP team owns, not hard-coded behaviour, and every change to them is audited.

Table 5. *Business rules*

Ref	Rule	Why
BR-01	Every document belongs to one of seven companies, determined from the sold-to name via a client-owned mapping. Unknown names are held for a person, never guessed.	Company drives the entity, tax, currency and approval route
BR-02	Documents from a marked prepaid vendor route to Prepayments regardless of how they look.	The vendor relationship overrides appearance
BR-03	A negative total or cancellation wording routes to Credits ahead of every other rule.	A negative document can never be posted as an invoice

Ref	Rule	Why
BR-04	A marked settlement vendor re-invoicing at a settled price routes to Rebills.	Only these vendors produce rebills
BR-05	A valid purchase order routes to PO invoices; everything else defaults to Non-PO.	The strongest positive signal wins; the safe default asks a person
BR-06	Below a configurable confidence threshold, a document waits in Manual Review; an unknown vendor lowers confidence.	Guessing is most dangerous exactly where confidence is low
BR-07	Duplicates are refused on vendor, invoice number and amount together.	Catches resends without blocking legitimate re-invoicing
BR-08	Net plus taxes plus freight must equal gross before a document is submitted.	An out-of-balance document must not reach Sage
BR-09	The two-way or three-way match type is read from the Sage purchase-order line.	The order defines the control, not the platform
BR-10	A receipt is identified uniquely by ship date and bill of lading; matching is receipt-level.	Makes thousand-receipt monthly summaries tractable
BR-11	The platform never creates a purchase order; a person attaches one or codes from a template.	PO creation is manual and cannot be automated
BR-12	A credit memo must reference the original purchase invoice it reverses.	Sage builds the credit from the original invoice's lines
BR-13	A rebill must name the purchase credit memo it settles, linking original, credit and rebill.	Closes the settlement chain that is unprovable today
BR-14	Every document is approved; an auto-matched document still needs approval unless AP exempts it.	Approval is a control, not a consequence of matching
BR-15	Approval routes by per-company limit with automatic escalation; nobody approves their own submission.	Enforced authority, not remembered authority
BR-16	AP final review fires on rules: amount, closed period, every credit, a new vendor, a released variance, or a named vendor.	Targets review where it adds control, not everywhere
BR-17	Writes to Sage are idempotent on the platform document number; a retry cannot post twice.	A timeout must never create a duplicate posting
BR-18	Nothing is deleted; cancellation and out-of-scope are states with reasons and stay searchable.	The record must always be auditable

8 Non-functional requirements

Requirements about how the platform must behave. Operational service levels, hosting and disaster recovery are commercial and delivery matters, covered in the Statement of Work.

Table 6. *Non-functional requirements*

Ref	Requirement	Target
NFR-1	Interactive response, queue lists, search, document open	Under 1.5 s at the 95th percentile at production volumes
NFR-2	Large document handling	A 1,000-line document opens, matches and submits whole; no line ceiling
NFR-3	Throughput	Sustained 3,500 documents a month, with true-up peaks near 3,000 more, without queue degradation
NFR-4	Extraction turnaround	Median under 2 minutes from arrival to extracted and classified
NFR-5	Authentication	Single sign-on with MFA enforced by policy; no local passwords
NFR-6	Write integrity	Idempotent Sage writes; a failed write surfaces as a visible state, never a silent loss
NFR-7	Authorisation	Role and company scoped; a user sees only the companies they are entitled to
NFR-8	Data protection	Encrypted in transit and at rest; secrets in a managed vault; no financial data in logs
NFR-9	Retention	Documents and audit retained 7 years by default, configurable per company
NFR-10	Accessibility	WCAG 2.1 AA, verified by automated audit
NFR-11	Browser support	Current Edge, Chrome, Firefox and Safari; usable on tablet for approvals

9 Acceptance criteria

The platform is accepted when each capability and document type behaves as below on production-representative data.

Table 7. *Acceptance criteria*

Area	Accepted when
Intake	A batch scan splits into separate, individually traceable documents, and a duplicate is refused naming the document that already holds the number.
Extraction	A 1,000-line document is read whole, and fields below the confidence threshold are flagged rather than accepted silently.
Classification	Sample documents route to the correct queue without correction, and a below-threshold document waits in Manual Review with the reason stated.
Validation	An out-of-balance document cannot be submitted, and a coding value not present in Sage is refused.

Area	Accepted when
PO matching	A three-way invoice with receipts present matches and codes with no human edit, and a price variance appears as an exception showing ordered versus invoiced.
Approval	An amount over an approver's limit escalates automatically, self-approval is blocked, and an email approval is recorded on the document.
Sage integration	A purchase invoice and a credit memo are created in Sage with the image attached and the number written back within seconds, and reconciliation flags any divergence.
Payment	The payment number and method are captured from Sage and the document is closed.
Credit and rebill	A credit cannot post without its original invoice reference, and a rebill chain either shows all three Sage numbers or states exactly which link is missing.
Search and reporting	Every displayed count clicks through to the exact documents behind it, and every view exports to spreadsheet.

10 Assumptions and dependencies

10.1 Assumptions

- Sage X3 exposes the required reads and document creation through a supported API; a file-based fallback would extend the integration effort.
- Extraction uses a managed document-intelligence service with a pre-trained invoice model; per-vendor custom models are not assumed.
- Configuration data, approval hierarchy, limits, templates, company and vendor mappings, is provided by H.J. Baker as data, not derived from documents.
- Volumes are approximately 3,300 invoices a month with quarterly true-up peaks, across seven companies, four currencies and roughly 2,700 vendors.
- Two Sage environments are available to the platform: one test and one production.
- Documents are English-language and supplied in the formats already shared.

10.2 Dependencies

Table 8. *Dependencies*

Ref	Dependency	Needed for
D-1	Net at Work engaged to confirm the Sage X3 API contract, plus a Sage test instance	Building the Sage integration to a firm contract
D-2	Read access to the AP mailbox and agreement on document routing	Intake and classification
D-3	Configuration data: hierarchy and limits, template library, company and vendor mappings, tolerances	Approvals, coding and classification, and UAT

Ref	Dependency	Needed for
D-4	Identity provider integration and a security review slot	Single sign-on and opening environments to users
D-5	Named AP-team testers for user acceptance testing	Acceptance sign-off
D-6	Answers to the open decisions in the Product Requirements Document	Removing rework risk from the build